

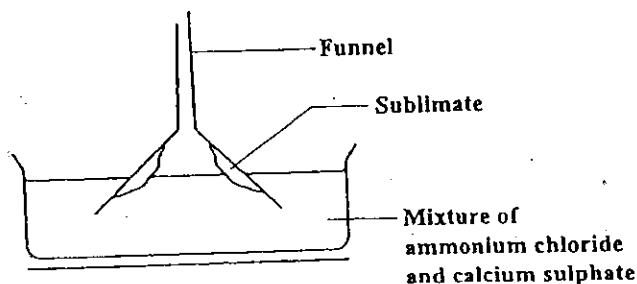
CSEC Chemistry June 2012

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1. Which of the following features is TRUE for the particles in a gas?

(A) Larger when heated
(B) Further apart when heated
(C) Very closely packed together
(D) Strong forces between them

Item 2 refers to the following experiment.



2. A mixture of ammonium chloride and calcium sulphate is heated, in the apparatus shown above, until no further change takes place. The sublimate would MOST probably be

(A) ammonium chloride
(B) calcium sulphate
(C) calcium sulphate and ammonium chloride
(D) calcium chloride and ammonium sulphate

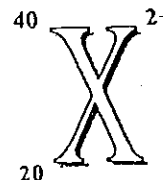
3. The atomic structures of X and Y are 2, 8, 5 and 2, 8, 6 respectively. Which of the following pairs of properties represents X and Y?

	X	Y
(A)	Metal	non-metal
(B)	Metal	metal
(C)	Non-metal	metal
(D)	Non-metal	non-metal

4. The mass number of an element is number of

(A) neutrons in an atom of the element
(B) protons and electrons in an atom of the element
(C) neutrons and protons in an atom of the element
(D) protons, neutrons and electrons in an atom of the element

5. In the diagram below, X represents symbol of a positive ion.



Which of the following represents number of neutrons and electrons in X

	Number of Neutrons	Number of Electrons
(A)	20	20
(B)	20	18
(C)	60	18
(D)	60	20

6. The arrangement of elements in the Periodic Table is based on

(A) atomic number
(B) mass number
(C) relative atomic mass
(D) relative molecular mass

7. The mass concentration of a potassium chloride solution is 60 g dm^{-3} . What is the mass of potassium chloride in 25 cm^3 of this solution?

(A) 0.0015 g
(B) 0.15 g
(C) 1.5 g
(D) 15.0 g

8. The quantity, '1 mole of atoms of an element', refers to the mass of

- (A) 1 atom of the element
- (B) the element which contains 6.0×10^{23} atoms
- (C) the element which occupies 24.0 dm^3 at STP
- (D) the element which combines completely with 12 g of carbon -12

9. Which of the following equations correctly represents the thermal decomposition of lead(II) nitrate?

- (A) $2\text{Pb}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{Pb}(\text{NO}_2)_2(\text{s}) + 2\text{O}_2(\text{g})$
- (B) $2\text{Pb}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{PbO}_2(\text{s}) + 4\text{NO}_2(\text{g})$
- (C) $2\text{Pb}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{PbO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$
- (D) $2\text{Pb}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{Pb}(\text{s}) + 4\text{NO}_2(\text{g}) + 2\text{O}_2(\text{g})$

10. Which of the following compounds has the GREATEST percentage by mass of carbon present in one mole? (M_r = relative molecular mass; relative atomic mass: H = 1; C = 12; O = 16)

- (A) $\text{C}_2\text{H}_4\text{O}_2$ ($M_r = 60$)
- (B) $\text{C}_6\text{H}_{12}\text{O}_6$ ($M_r = 180$)
- (C) $\text{C}_2\text{H}_6\text{O}$ ($M_r = 46$)
- (D) $\text{C}_3\text{H}_8\text{O}$ ($M_r = 60$)

Items 11–12 refer to Compound X, which melts at 18°C and boils at 55°C .

11. Which of the following statements is true about the intermolecular forces of attraction in Compound X?

- (A) They are strong.
- (B) They increase above 18°C .
- (C) They decrease above 18°C .
- (D) They increase above 55°C .

12. Compound X is MOST likely to be

- (A) a giant molecule
- (B) a liquid metal
- (C) a simple molecule
- (D) an ionic compound

13. Which of the following elements can exist in more than one form in the SAME state?

- (A) Carbon
- (B) Iron
- (C) Iodine
- (D) Neon

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14. Graphite can be used as a lubricant because of the

- (A) loose electrons which can move throughout the lattice
- (B) weak attraction among the hexagonal layers of carbon atoms
- (C) strong attraction among the hexagonal layers of carbon atoms
- (D) strong attraction within the hexagonal layers of carbon atoms

15. Sublimation can be used in separating

- (A) suspended solids from a liquid
- (B) a pair of liquids that do not mix
- (C) components of a liquid with similar boiling points
- (D) a solid with weak interparticle forces from one with strong forces

16. Which of the following substances would be expected to show the LEAST solubility in water?

- (A) Ammonia
- (B) Ethanol
- (C) Iodine
- (D) Lithium chloride

17. Silicon (Si) is in the same group of the Periodic Table as carbon. The compound formed between silicon and hydrogen is likely to have the formula

- (A) SiH_2
- (B) SiH_4
- (C) Si_2H_3
- (D) Si_3H_5

18. Barium (Ba) is an element in Group II of the Periodic Table. It reacts with cold water to give hydrogen and another substance of formula

- (A) BaO
- (B) Ba_2O
- (C) BaOH
- (D) Ba(OH)_2

19. Which of the following is NOT true of the group of elements known as the halogens?

- (A) They are all non-metals.
- (B) They form negative ions by the loss of electrons from their atoms.
- (C) The boiling point of the elements increases as their atomic numbers increase.
- (D) The oxidizing power of the elements decreases as their atomic numbers increase.

Item 20 refers to one mole of EACH of the following acids.

- I. H_2SO_4
- II. CH_3COOH
- III. $(\text{COOH})_2$

20. Which of the above acids require more than one mole of NaOH(aq) for complete neutralization?

- (A) II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

21. Which of the following terms BEST describes strong acids?

- (A) Concentrated
- (B) Partially ionized
- (C) Ionized to a large extent
- (D) Concentrated and partially ionized

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22. A solution, when treated with the gas sulphur dioxide, becomes green. The solution contains potassium
- (A) nitrate
(B) sulphate
(C) dichromate(VI)
(D) manganate(VII)
23. Which of the following observations would be expected when aqueous sodium hydroxide is added to an ammonium salt and the mixture is warmed?
- (A) Alkaline gas produced
(B) White precipitate soluble in excess
(C) White precipitate insoluble in excess
(D) Green precipitate insoluble in excess
24. Which of the following pairs of substances would be MOST suitable for the preparation of copper sulphate crystals?
- (A) Copper and dilute sulphuric acid
(B) Copper and concentrated sulphuric acid
(C) Copper nitrate and dilute sulphuric acid
(D) Copper oxide and dilute sulphuric acid
25. When chlorine reacts with sodium to form sodium chloride, the chlorine is acting as
- (A) a base
(B) an acid
(C) a reducing agent
(D) an oxidizing agent
26. What is the change in the oxidation number of nitrogen in the equation below?
- $$4 \text{NH}_3(\text{g}) + 5 \text{O}_2(\text{g}) \rightarrow 4 \text{NO}(\text{g}) + 6 \text{H}_2\text{O}(\text{g})$$
- (A) + 3 to + 2
(B) + 3 to - 2
(C) - 3 to + 2
(D) - 3 to - 2
27. Which of the following reactions would you expect to take place?
- I. $\text{Zn}(\text{s}) + \text{Mg}^{2+}(\text{aq}) \rightarrow \text{Mg}(\text{s}) + \text{Zn}^{2+}(\text{aq})$
II. $\text{Mg}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Cu}(\text{s}) + \text{Mg}^{2+}(\text{aq})$
III. $\text{Cu}(\text{s}) + \text{Mg}^{2+}(\text{aq}) \rightarrow \text{Mg}(\text{s}) + \text{Cu}^{2+}(\text{aq})$
- (A) I only
(B) II only
(C) I and III only
(D) II and III only
- Items 28–29 refer to the following acids.
- (A) Sulphuric acid
(B) Hydrochloric acid
(C) Nitric acid
(D) Ethanoic acid
- Match EACH item below with ONE of the options above. EACH option may be used more than once, once or not at all.
- Which of the acids above
28. has a basicity of 2?
29. is weakly ionised in aqueous solutions?

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30. Which of the following is a weak electrolyte?

- (A) Molten lead
- (B) Molten lead bromide
- (C) Aqueous ethanoic acid
- (D) Dilute hydrochloric acid

31. During the electrolysis of aqueous copper(II) sulphate solution using inert electrodes, the ions migrating to the cathode are

- (A) $\text{Cu}^{2+}(\text{aq})$ and $\text{SO}_4^{2-}(\text{aq})$
- (B) $\text{H}^+(\text{aq})$ and $\text{OH}^-(\text{aq})$
- (C) $\text{SO}_4^{2-}(\text{aq})$ and $\text{OH}^-(\text{aq})$
- (D) $\text{Cu}^{2+}(\text{aq})$ and $\text{H}^+(\text{aq})$

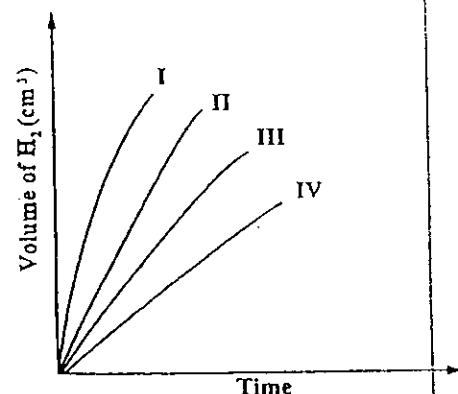
32. In the anodizing of an aluminium pot, which of the following is TRUE?

- (A) Hydrogen is given off at the anode.
- (B) A layer of aluminium hydroxide forms on the pot.
- (C) The aluminium pot is the anode in the cell.
- (D) The electrolyte is a solution of sodium chloride.

33. How many coulombs (C) of electricity passes through an electrolyte when a current of 100 A flows for 10 minutes?

- (A) 100
- (B) 600
- (C) 10 000
- (D) 60 000

Item 34 refers to the graph below which shows the rate of reaction for different particle sizes of the same mass of magnesium reacting with dilute acid.



34. Which line on the graph, I, II, III or IV, would BEST represent the reaction with powdered magnesium?

- (A) I
- (B) II
- (C) III
- (D) IV

35. Which of the following statements is true about the reactants and products in an exothermic reaction?

- (A) Products have more stored energy than the reactants.
- (B) Reactants have more stored energy than the products.
- (C) Products and reactants have the same amount of stored energy.
- (D) Reactants have less energy than the products since energy is absorbed from the surroundings.

36. When crystals of potassium nitrate are dissolved in water, the temperature of the solution decreases because

(A) energy is required to break down the crystal structure of the potassium nitrate
 (B) heat is always absorbed when a substance dissolves
 (C) the water absorbs the energy from the potassium nitrate
 (D) the energy content of dissolved potassium nitrate is lower than that of solid potassium nitrate

37. In which of the following industrial preparations is a catalyst NOT used?

I. Sulphuric acid
 II. Chlorine
 III. Ammonia

(A) I only
 (B) II only
 (C) I and II only
 (D) II and III only

38. A metal, M, can be extracted by roasting its ore in air with carbon. Metal M could be

(A) Mg
 (B) Al
 (C) Ca
 (D) Fe

39. Which of the following metals can be extracted by electrolysis?

(A) K
 (B) Ca
 (C) Al
 (D) Fe

Items 40–42 refer to the properties of four metals, W, X, Y and Z, which are summarised in the following table.

Metal	Reaction With Water/Steam
W	There is no reaction.
X	Reacts with steam but not with cold water.
Y	Reacts rapidly with cold water.
Z	Reacts slowly with cold water.

40. Which of the metals is NOT likely to react with dilute hydrochloric acid to give hydrogen?

(A) W
 (B) X
 (C) Y
 (D) Z

41. Which of the metals is likely to form a soluble hydroxide?

(A) W
 (B) X
 (C) Y
 (D) Z

42. For which of the following pairs of metals is the first MOST likely to displace the second solution?

(A) X, Y
 (B) Z, X
 (C) W, X
 (D) Z, Y

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43. Which of the following compounds will NOT evolve carbon dioxide upon heating?

(A) Calcium carbonate
(B) Copper(II) carbonate
(C) Lead(II) carbonate
(D) Sodium carbonate

44. Which of the following observations is likely when EXCESS aqueous ammonia is added to a solution of copper(II) sulphate and the mixture is shaken?

(A) Green solution
(B) Green precipitate
(C) Deep blue solution
(D) Light blue precipitate

45. Which of the following are the raw materials and conditions required for the fermentation of sugars?

Raw Materials	Temperature	pH
(A) Glucose + yeast	room temperature	7
(B) Starch + yeast	30 °C above room temperature	7
(C) Sucrose + yeast	30 °C above room temperature	less than 7
(D) Sucrose + glucose	room temperature	less than 7

46. During the digestion of polysaccharides by the human body, the main process occurring is

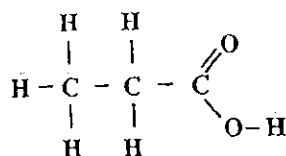
(A) condensation
(B) hydrolysis
(C) neutralisation
(D) polymerisation

Items 48–49 refer to the following options

(A) C_2H_4
(B) C_2H_6
(C) $CH_3(CH_2)_2COOH$
(D) $CH_3(CH_2)_3COH$

Match EACH item below with ONE of the options above. Each option may be used more than once, once or not at all.

47. To which homologous series does the following structure belong?



(A) Acids
(B) Alkanes
(C) Alkenes
(D) Alcohols

48. An alcohol

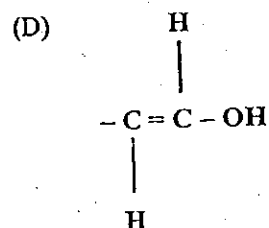
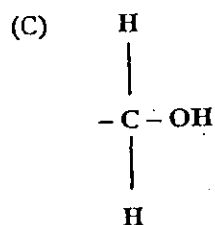
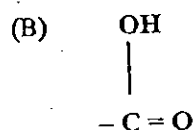
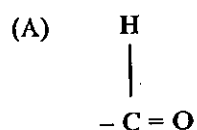
49. An alkane

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50. The compound ethene is described as being unsaturated. This means that the

(A) carbon atoms in ethene are linked by single bonds
(B) carbon atoms in the molecule are very reactive
(C) molecule has insufficient hydrogen atoms
(D) molecule contains at least one double bond

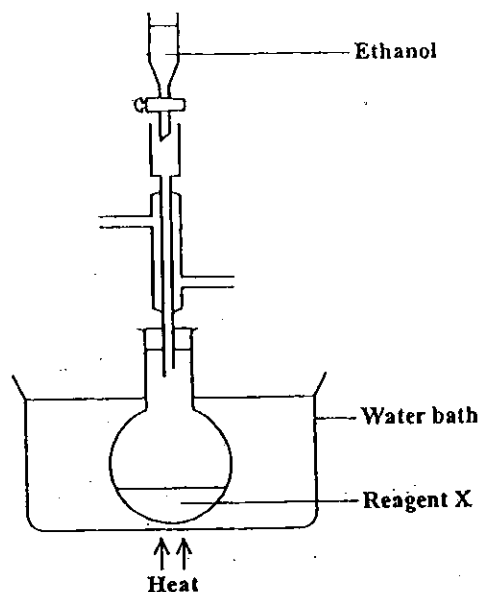
51. The functional group present in carboxylic acids is



52. When ethanoic acid and ethanol react to form ethyl ethanoate, the catalyst used is

(A) dilute sulphuric acid
(B) dilute hydrochloric acid
(C) concentrated hydrochloric acid
(D) concentrated sulphuric acid

53. Ethanol can be converted to ethanoic acid using the apparatus shown below.



Reagent X could be

(A) aqueous bromine
(B) concentrated sulphuric acid
(C) acidified potassium dichromate(VI)
(D) aqueous sodium hydroxide

Items 54–55 refer to the following compounds.

(A) Fat
(B) Starch
(C) Protein
(D) Polyester

Match EACH item below with ONE of the options above. Each option may be used more than once, once or not at all.

54. Which compound produces amino-acids when hydrolysed?
55. Which compound is NOT considered to be a polymer?

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56. A polyamide is formed by a condensation reaction between a molecule containing

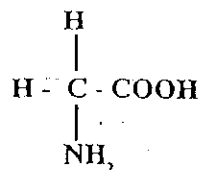
- (A) at least one -OH group with another containing at least one -NH₂ group
- (B) at least one -OH group with another containing at least one -COOH group
- (C) at least one -COOH group with another containing at least one -NH₂ group
- (D) two -OH groups with another containing two -COOH groups

57. Which of the following statements about polymers are true?

- I. Polymers are long-chain molecules made by joining together a large number of monomer units.
- II. Condensation polymers are generally made up from one type of monomer.
- III. Addition polymerisation involves the breaking of a C = C double bond in the monomer.

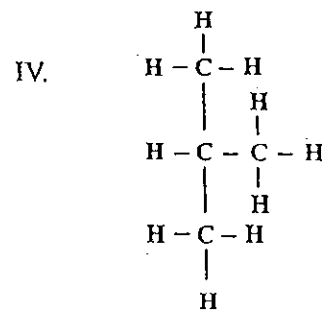
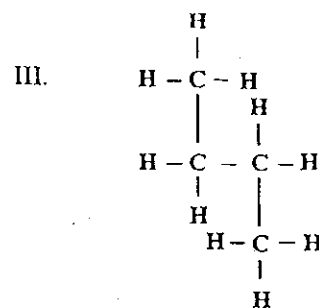
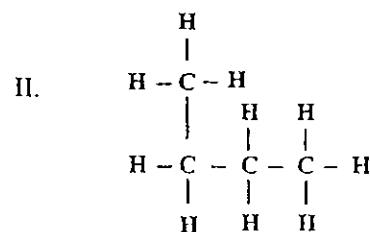
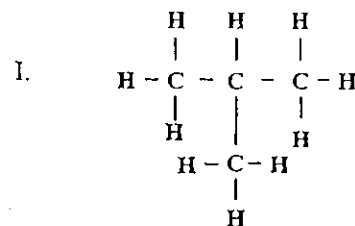
- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

58. Which of the following types of polymer may be derived from compounds like the one below?



- (A) Polyamide
- (B) Polyalkene
- (C) Polyester
- (D) Polysaccharide

Item 59 refers to the following structures

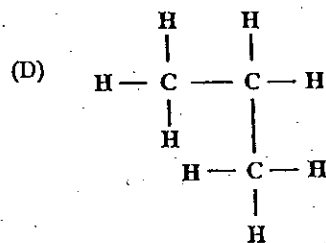
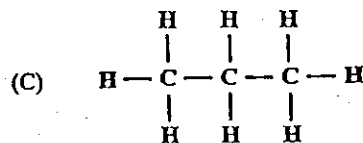
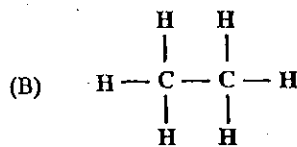
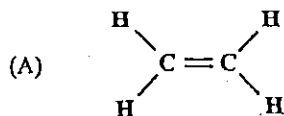


59. Which of the structures above represent branched alkanes?

- (A) I and II only
- (B) I and IV only
- (C) III and IV only
- (D) I, II and III only

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60. Which of the following compounds can decolourize potassium manganate(VII)?



END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.